

FHENet: Lightweight Feature Hierarchical Exploration Network for Real-Time Rail Surface Defect Inspection in RGB-D Images

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Abstract—In recent years, computer vision systems have been increasingly applied to rail defect inspection. Rail defects should be identified quickly and accurately to ensure safe, stable, and fast train operations and thereby reduce the incidence of accidents and economic losses. As most existing methods focus on accuracy, they cannot be deployed on mobile devices with limited computational resources. To solve this problem, we propose a lightweight feature hierarchical exploration network (FHENet) for real-time rail surface defect inspection. First, boundary textures of rail defects are acquired based on the maximum function and maximum pooling in a novel boundary extraction module (BEM), which improves boundary prediction while avoiding heavy computations. Second, a novel cross-modality exploration module (CEM) complements prominent regions through basic operations to avoid complex inference while providing high detection performance. Third, a novel multifeature integration module (MIM) optimizes representative feature regions by using simple operations to avoid complex computations. Results from extensive experiments demonstrate the superiority of the proposed FHENet to 14 state-of-the-art methods. Regarding efficiency, FHENet outperforms the state-of-the-art methods with only 5.26 M parameters and a processing speed of 60.33 frames/s. The FHENet code and results are available at <https://github.com/hjklearn/Rail-Defect-Detection>.

Index Terms—Boundary extraction, cross-modality exploration, feature hierarchical exploration network (FHENet), multifeature integration, rail defect inspection.

I. INTRODUCTION

WITH the gradual improvement in the load and speed of bullet trains, railways face increasing operational pressure [1] that may damage infrastructure and shorten service life. For example, defects (e.g., holes and cracks) appearing on a rail surface must be timely detected to prevent accidents [2], [3], [4].

Conventional rail defect detection generally depends on the work of experienced inspectors, thus requiring vast human resources and long working cycles. Alternatively, automated methods, such as those based on convolutional neural networks, are being widely studied for rail surface

defect inspection because of the rapid advancement of deep learning. Accordingly, various methods for rail defect inspection based on deep learning have recently been proposed. Liu et al. [5] proposed a network for rail defect detection in red–green–blue (RGB) images that includes a pyramid feature extraction module in a convolutional neural network. Zhang et al. [6] designed a multimodel convolutional neural network for rail defect inspection that includes a modified single-shot multibox detector, and you only look once with algorithm version 3. Yang et al. [7] proposed a method based on deep learning to perform multilevel end-to-end rail defect inspection. Liu and He [8] designed an end-to-end detection network based on triple attention for defect detection. These methods consider a single-modality input to achieve high detection performance of rail defects. However, these methods often provide unsatisfactory detection results when the color and texture of rail defects are similar to those of the background.

As surface defects in depth images indicate clear differences from the background, RGB-depth (RGB-D) rail surface defect inspection is receiving special attention [9]. In this sense, we can note two problems: 1) RGB rail defect detection may fail to achieve satisfactory results in complex scenes and 2) RGB-D rail defect detection generally involves numerous parameters and complex computations that are inapplicable to devices with limited computational resources. To solve these problems, we propose a feature hierarchical exploration network (FHENet) that balances inspection time and accuracy. Specifically, a lightweight MobileNetV2 backbone [10] extracts RGB and depth features. In addition, a novel boundary extraction module (BEM) provides clear boundaries related to complex rail defects. Moreover, we introduce a hierarchical exploration strategy comprising two key components, namely, a cross-modality exploration module (CEM) and a multifeature integration module (MIM), to find remarkable and complementary defect regions in RGB and depth data.

The main contributions of the proposed FHENet are summarized as follows:

1) The end-to-end lightweight FHENet for rail defect inspection is the first multimodality real-time lightweight method of its kind, achieving 60 frames/s processing speed with only 5.26 M parameters.

2) To acquire distinct boundary features, the BEM is based on maximum function and maximum pooling, clearly decreasing the computational cost and accelerating inference.

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3) Considering a lightweight implementation, a simple hierarchical exploration strategy deals with multimodality and multilevel feature integration. The strategy adopts shortcuts (i.e., addition and concatenation) for hierarchical processing, and the CEM and MIM highlight consistent target features and discard unrepresentative features.

II. RELATED WORK

A. Rail Surface Defect Inspection

Given the sustained development of deep learning, the detection performance of rail defects is prominently improving. Dong et al. [11] designed a multiple supervision framework that processes multiscale information to perform defect inspection. Song et al. [12] investigated surface defect detection using an encoder–decoder residual (EDR) network to achieve fine detection results. Zhang et al. [13] proposed an end-to-end multilevel segmentation network for rail defect detection. Zhang et al. [14] proposed two learning structures, separately applied them to represent common/heavy rail defects, and trained the network parameters using line-level labels. Wang et al. [9] designed a multimodality network (CLA) based on RGB-D images for rail defect detection; the network contains a multimodality attention block highlighting defects and a dual-stream decoder to mitigate noise. Zhou et al. [15] proposed an edge-aware multilevel interactive network (EMI) with feature integration and edge optimization for surface defect detection. Han et al. [16] designed a two-stage edge reuse network for strip steel surface defect inspection, comprising prediction and refinement stages. Ding et al. [17] proposed a framework (CSEP) for surface defect detection to explore the relations between features at different scales. Zhou et al. [18] designed an end-to-end network (DAC) for surface defect detection using an attention-guided cascade to process features for prediction.

Although existing methods adequately perform rail defect inspection, they neglect real-time operation. Meanwhile, most methods use single-stream encoding but neglect information such as depth maps to improve defect detection. Overall, a lightweight and accurate rail defect detection method should be devised to ensure safe train operations.

B. RGB-D Target Detection

RGB and depth data can be combined to improve target detection. Early target detection methods generally depend on handcrafted features for inspection [19], [20], [21], [22], [23], [24], [25], [26], [27], [28]. For example, Ju et al. [19] proposed a network for target detection using anisotropic center–surround differences (ACSD) in the depth image and considered depth and location priors to refine prediction. Cong et al. [20] designed a framework based on depth confidence (DCMC) to detect salient regions in stereoscopic images. Qu et al. [21] introduced a convolutional neural network (DF) to integrate cross-modality features for detection. Zhu et al. [22] used center-dark channel priors (CDCP) of images for target detection. These methods have demonstrated that correct feature extraction is important to improve detection performance.

With the emergence of deep learning, several methods based on high-performance encoders have been proposed to further enhance predictions. Piao et al. [23] developed an end-to-end depth-induced network (DMRA) for object detection by adopting a recurrent mechanism to obtain high-quality predictions. Wang and Gong [24] proposed a network (S2MA) for target detection with a dual-stream encoder to extract features and a saliency fusion module for adaptive predictions. Liu et al. [25] introduced fusion attention (CONET) with learning from different modalities to perform saliency detection. Ji et al. [26] designed a collaborative learning strategy to explore relations between cross-modality features. Li et al. [27] proposed a network (HAI) for saliency detection by adopting a hierarchical structure to handle feature interactions. Zhai et al. [28] designed a network (BBS) to complete saliency detection with a depth-enhanced module and cascaded decoder.

The aforementioned methods show accurate detection in diverse scenes, but they often neglect the computational cost and number of model parameters, making them unsuitable for deployment in mobile devices. By contrast, the proposed FHENet for rail defect inspection focuses on both detection accuracy and lightweight implementation for real-time operation.

III. PROPOSED FHENET

A. Overview

The architecture of the proposed FHENet is shown in Fig. 1. The MobileNetV2 backbone [10], denoted as $En-i$, is used to extract features R_i and D_i ($i = 1, 2, 3, 4, 5$) from RGB and depth data, respectively. The BEM is applied to low levels to acquire rail defect boundary features $E_{g,3}$. To obtain the most representative semantic features of defect regions, we introduce an efficient hierarchical exploration strategy that comprises the simple CEM and MIM and enables lightweight implementation. The outputs of every stage throughout hierarchical exploration are $E_{r/d,i}$ ($i = 2, 3, 4, 5$) and S_i ($i = 1, 2, 3$). In the first stage, to compensate for the loss of original features, we add M_j to both R_i and D_i ($i \neq 1$). The second stage of boundary feature enhancement, denoted as U, further optimizes boundary features, providing outputs $E_{g,2}$ and $E_{g,1}$. Deep and boundary supervision processes are applied for accelerating and stabilizing training.

B. Boundary Extraction Module

Low-level features generally contain boundary and texture information to promote rail defect detection, but they also contain noise. To mitigate noise and obtain clear boundaries for rail defects, we introduce the BEM architecture shown in Fig. 2. First, considering features R_1 and D_1 , we adopt 1×1 convolutions to increase the number of channels and thereby improve the boundary features at the expense of increased computational cost. Then, addition and multiplication are used to achieve cross-modality fusion as follows:

$$f_1 = (C_{1 \times 1}(R_1) \otimes C_{1 \times 1}(D_1)) \oplus (C_{1 \times 1}(R_1) \oplus C_{1 \times 1}(D_1)) \quad (1)$$

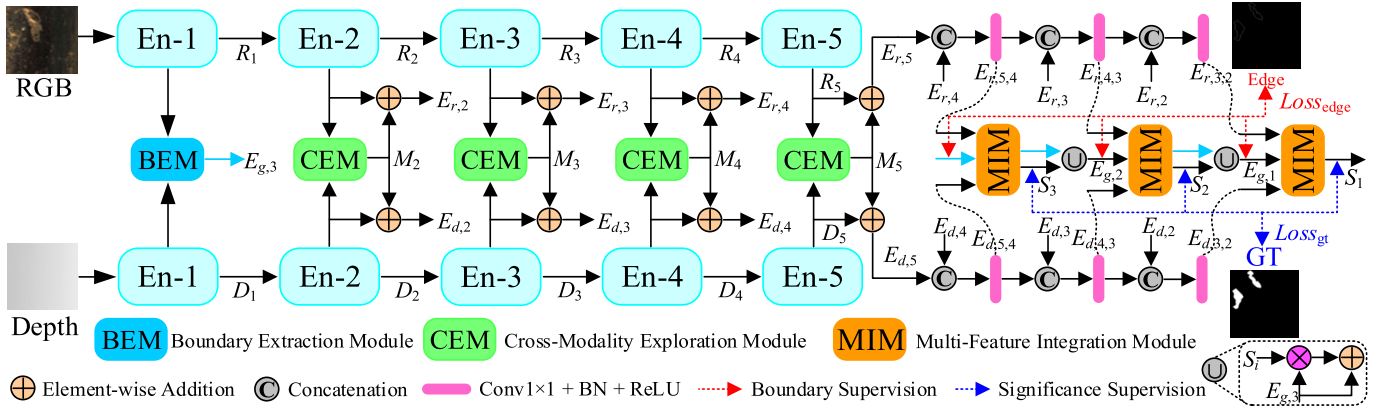


Fig. 1. Architecture of proposed FHENet. (BN, batch normalization; Conv, convolution; GT, ground truth, ReLU, rectified linear unit.)

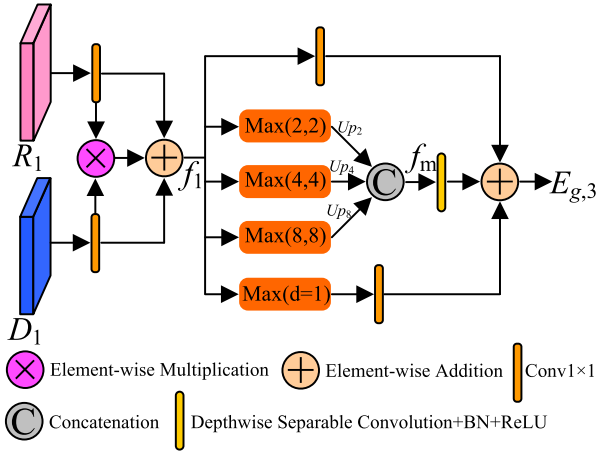


Fig. 2. Architecture of BEM.

where $C_{1 \times 1}$, $\oplus$, and $\otimes$ denote a 1×1 convolutional layer, elementwise addition, and elementwise multiplication, respectively.

To avoid a complex inference, we apply depth-wise separable convolution, maximum function, and maximum pooling to extract boundary features. We use a residual connection to avoid information loss for enhanced feature expression. Finally, the results are added to obtain boundary output $E_{g,3}$. The BEM is formulated as follows:

$$f_m = \text{Cat}(\text{Up}_2(\text{Max}_2(f_1)), \text{Up}_4(\text{Max}_4(f_1)), \text{Up}_8(\text{Max}_8(f_1))) \quad (2)$$

$$E_{g,3} = C_{1 \times 1}(f_1) \oplus \text{DC}_{3 \times 3}(f_m) \oplus C_{1 \times 1}(\text{Max}_{d=1}(f_1)) \quad (3)$$

where Cat , Up_i , $\text{DC}_{3 \times 3}$, $\text{Max}_{d=1}$, and Max_i denote concatenation along the channel axis, bilinear interpolation, 3×3 depth-wise separable convolution with batch normalization and rectified linear unit activation, maximum pixel value across channels, and maximum pooling, respectively.

C. Cross-Modality Exploration Module

The lightweight backbone generates features that contain scarce representative information because it has few convolutional layers and learnable parameters. Similar to the method in [27], we introduce the CEM architecture shown in

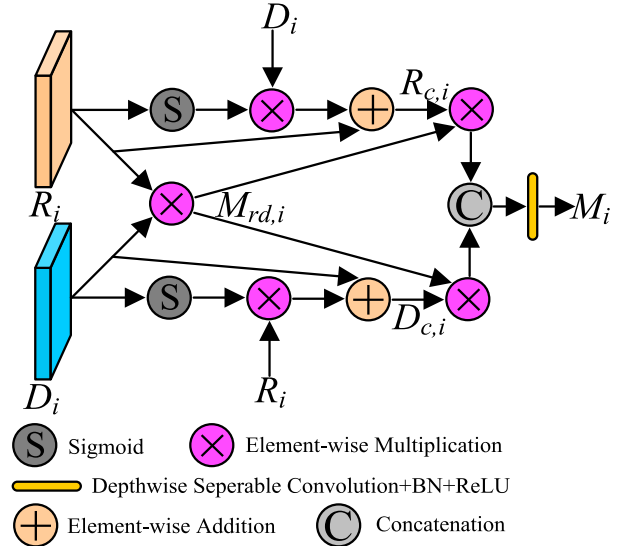


Fig. 3. Architecture of CEM.

Fig. 3 to compensate for deficient features in the lightweight network. First, feature weights of R_i and D_i ($i = 2, 3, 4, 5$) are obtained by applying the sigmoid function and multiplying the weights by the complementary features to obtain cross-modality weighted feature interactions. As these operations can cause information loss, we use a residual connection to preserve the original features and obtain features $R_{c,i}$ and $D_{c,i}$ as follows:

$$R_{c,i} = R_i \oplus (\text{Sig}(R_i) \otimes D_i) \quad i = 2, \dots, 5 \quad (4)$$

$$D_{c,i} = D_i \oplus (\text{Sig}(D_i) \otimes R_i) \quad i = 2, \dots, 5 \quad (5)$$

where Sig represents the sigmoid function.

Next, we multiply R_i with D_i to highlight common features and separately multiply the result with $R_{c,i}$ and $D_{c,i}$ to obtain common cross-modality fusion features. Finally, we concatenate the features along the channel axis, and a depth-wise separable convolution optimizes the features and reduces the computational cost. The CEM is formulated as follows:

$$M_{rd,i} = R_i \otimes D_i \quad i = 2, \dots, 5 \quad (6)$$

$$M_i = \text{DC}_{3 \times 3}(\text{Cat}(M_{rd,i} \otimes R_{c,i}, M_{rd,i} \otimes D_{c,i})) \quad i = 2, \dots, 5. \quad (7)$$

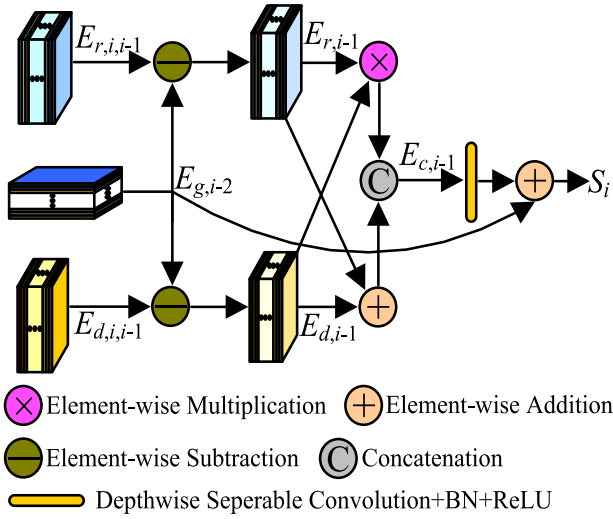


Fig. 4. Architecture of MIM.

D. Multifeature Integration Module

The aforementioned processes provide cross-modality fusion and boundary features. Considering a lightweight architecture, we introduce the simple MIM architecture shown in Fig. 4 to integrate features. Specifically, aiming to remove redundant information, the feature $E_{g,i-2}$ ($i = 3, 4, 5$) is separately subtracted from $E_{d,i,i-1}$ and $E_{r,i,i-1}$ ($i = 3, 4, 5$) to obtain $E_{d,i-1}$ and $E_{r,i-1}$, respectively. Then, multiplication and addition are used to fuse features $E_{d,i-1}$ and $E_{r,i-1}$, and concatenation is applied to obtain feature $E_{c,i-1}$, while a depth-wise separable convolution optimizes the confluent features. Finally, feature $E_{g,i-2}$ is added to include boundary information. The MIM is formulated as follows:

$$E_{r,i-1} = |E_{r,i,i-1} - E_{g,i-2}| \quad i = 3, 4, 5 \quad (8)$$

$$E_{d,i-1} = |E_{d,i,i-1} - E_{g,i-2}| \quad i = 3, 4, 5 \quad (9)$$

$$E_{c,i-1} = \text{Cat}(E_{r,i-1} \otimes E_{d,i-1}, E_{r,i-1} \oplus E_{d,i-1}) \quad i = 3, 4, 5 \quad (10)$$

$$S_{i-2} = \text{DC}_{3 \times 3}(E_{c,i-1}) + E_{g,i-2} \quad i = 3, 4, 5 \quad (11)$$

where S_{i-2} denotes the resulting defect detection with index ($i - 2$).

E. Supervision

We use deep supervision and boundary supervision with hybrid loss to train the proposed FHENet. Let G , S_{i-2} , and $E_{g,i-2}$ represent the ground truth, defect prediction, and boundary prediction, respectively. The total loss is given by

$$L_{\text{total}} = \sum_{i=3}^5 \left[\text{BCE}(S_{i-2}, G) + \text{IOU}(S_{i-2}, G) + \text{BCE}(E_{g,i-2}, G) + \text{IOU}(E_{g,i-2}, G) \right] \quad (12)$$

where BCE and IOU denote binary cross-entropy and intersection over union, respectively.

IV. EXPERIMENTS AND RESULTS

A. Dataset and Evaluation Metrics

The NEU RSSDS-AUG dataset collected 15 P60 rail tracks between 130 and 400 mm in length by binocular stereovision,

Algorithm 1 FHENet Training

Input: NEU RSSDS-AUG data(images, ground truth—GT, boundary)

Output: M_weights of trained FHENet

batch_{train} $\leftarrow$ 4, epochs $\leftarrow$ 150, total_mae $\leftarrow$ 0, best_mae $\leftarrow$ 1

$D_{\text{train}} \leftarrow$ NEU RSSDS-AUG data(images, GT, batch_{train}, epochs)

for epoch **in** range(epochs)

for (images, GT, boundary) **in** enumerate(D_{train})

S_{i-2} , $E_{g,i-2}$ ($i = 3, 4, 5$) $\leftarrow$ FHENet(images, GT)

$L_{\text{total}} \leftarrow$ calculate loss using Eq. (12)

weight $\leftarrow$ update($\partial L_{\text{total}} / \partial \text{weight}$)

train_mae $\leftarrow$ metric(S_1 , GT)

total_mae $\leftarrow$ train_mae + total_mae

mae = train_mae/len(D_{train})

if mae < best_mae

best_mae $\leftarrow$ mae

save_net(M_weights)

return M_weights

resulting in a total of 113 images with high pixel values. To adequately apply deep learning to this dataset, a variety of random grid methods were employed to augment the data. A more detailed description of the dataset is available in [9]. We used 1500 and 362 RGB-D image pairs from the dataset to construct the training and test sets, respectively. To evaluate the performance of the proposed FHENet, we adopted nine common metrics of target detection [28], [29], [30], [31], [32], [33], [34], [35], [36], including the precision-recall (P-R) curve, precision (Pre), recall (Rec), F1-score (F1), intersection over union (IOU), maximum E-measure (maxEm), S-measure (Sm), maximum F-measure (maxFm), and mean absolute error (MAE) [32].

B. Implementation Details

The proposed FHENet was implemented using the PyTorch framework, and training proceeded as described in Algorithm 1. Adam optimization was applied to the proposed model. The initial learning rate was set to $1e - 4$ and then divided by 2 every 20 epochs, and the batch size was set to 4. The training and test RGB-D images were resized to 320×320 pixels. Training proceeded for 150 epochs. During testing, the inference time of the final prediction was 0.016 s (60.33 frames/s) and 0.110 s (9.05 frames/s) in computers equipped with an NVIDIA TITAN Xp GPU and Intel i5-7500 CPU, respectively.

C. Performance Evaluation

We compared the proposed FHENet with 14 state-of-the-art target detection methods: ACSF [19], DCMC [20], DF [21], CDCP [22], DMRA [23], S2MA [24], CONET [25], EMI [15], CSEP [17], EDR [12], DAC [18], HAI [27], BBS [28], and CLA [9].

Table I lists the quantitative results of four evaluation metrics for the methods on the test set. The proposed FHENet clearly outperforms the comparison methods in all the metrics. For example, FHENet achieves improvements of 7.2% for MAE, from 0.920 to 0.926 for maxEm, and from 0.878 to

TABLE I
EVALUATION METRICS OBTAINED FROM COMPARED METHODS. THE
BEST RESULTS ARE SHOWN IN BOLD

Models	NEU RSDDS-AUG			
	$S_m \uparrow$	maxEm $\uparrow$	maxFm $\uparrow$	MAE $\downarrow$
DCMC	0.484	0.595	0.498	0.287
ACSD	0.556	0.670	0.575	0.360
DF	0.564	0.713	0.636	0.241
CDCP	0.574	0.694	0.591	0.236
DMRA	0.736	0.834	0.783	0.141
HAI	0.718	0.829	0.803	0.171
S2MA	0.775	0.864	0.817	0.141
CONET	0.786	0.878	0.834	0.101
EMI	0.800	0.876	0.850	0.104
CSEP	0.814	0.899	0.866	0.085
EDR	0.811	0.893	0.850	0.082
BBS	0.828	0.909	0.867	0.074
DAC	0.824	0.911	0.875	0.071
CLA	0.835	0.920	0.878	0.069
Ours	0.836	0.926	0.881	0.064

TABLE II
TEST RESULTS (PRE, REC, $F1$, AND IOU) OF THE PERFORMANCE OF THE
RELEVANT METHODS. THE BEST RESULTS ARE SHOWN IN BOLD

Models	Pre	Rec	$F1$	IOU
DCMC	66.16%	25.46%	33.36%	19.23%
ACSD	55.93%	63.88%	56.65%	40.63%
DF	78.88%	31.02%	42.12%	22.41%
CDCP	73.07%	36.14%	44.98%	27.86%
DMRA	80.36%	74.18%	74.84%	62.96%
HAI	73.90%	91.67%	78.98%	68.91%
S2MA	76.91%	82.83%	78.20%	70.39%
CONET	86.85%	78.61%	80.55%	70.57%
EMI	82.65%	87.76%	83.31%	74.82%
CSEP	85.29%	87.61%	85.14%	76.65%
EDR	85.32%	86.60%	84.12%	75.39%
BBS	86.27%	87.31%	85.63%	77.27%
DAC	86.71%	88.09%	86.23%	77.77%
CLA	87.27%	86.59%	86.07%	77.87%
Ours	87.22%	88.34%	87.01%	78.93%

0.881 for maxFm compared with the second-best method. Table II shows the quantitative results of Pre, Rec, $F1$, and

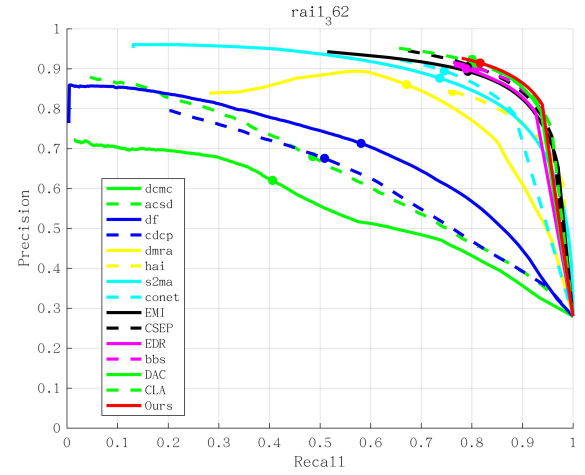


Fig. 5. P-R curves of evaluated methods.

TABLE III
QUANTITATIVE COMPARISON OF EVALUATED METHODS. THE BEST
RESULTS ARE SHOWN IN BOLD

Models	FLOPs/G $\downarrow$	Params/M $\downarrow$	Size/MB $\downarrow$	fps(gpu) $\uparrow$	fps(cpu) $\uparrow$
DMRA	241.728	59.65	238.9	27.85	0.85
HAI	73.465	59.82	239.7	13.41	0.81
S2MA	108.060	86.65	346.9	30.17	0.84
CONET	27.400	43.66	167.6	51.71	2.23
EMI	106.86	99.13	396.8	35.70	1.04
CSEP	45.43	18.78	75.5	28.31	1.39
EDR	32.24	39.31	157.6	30.41	1.87
BBS	12.690	49.78	200.1	40.22	2.74
DAC	109.28	98.39	393.9	28.31	0.94
AFNET	130.696	31.44	125.9	51.02	0.77
CLA	82.568	184.15	738.2	16.76	1.98
Ours	2.714	5.26	21.6	60.33	9.05

IOU for each method. The proposed FHENet achieves a higher performance regarding IOU. Although our model is not the best in terms of Pre and Rec compared with the other methods, it ranks second. In addition, $F1$ provides a measure of overall performance, with the proposed FHENet achieving the best result, indicating its competitiveness with respect to other methods. The P-R curves of the evaluated methods are shown in Fig. 5.

We quantitatively compared FHENet with various state-of-the-art methods regarding the number of floating-point operations (FLOPs), run time, model size, and the number of parameters and obtained the results listed in Table III. We used the NVIDIA TITAN Xp GPU for the evaluation of all the methods to ensure a fair comparison. Compared with the CLA model, the frames per second of our proposed model are more than three times faster than the CLA model, the model size

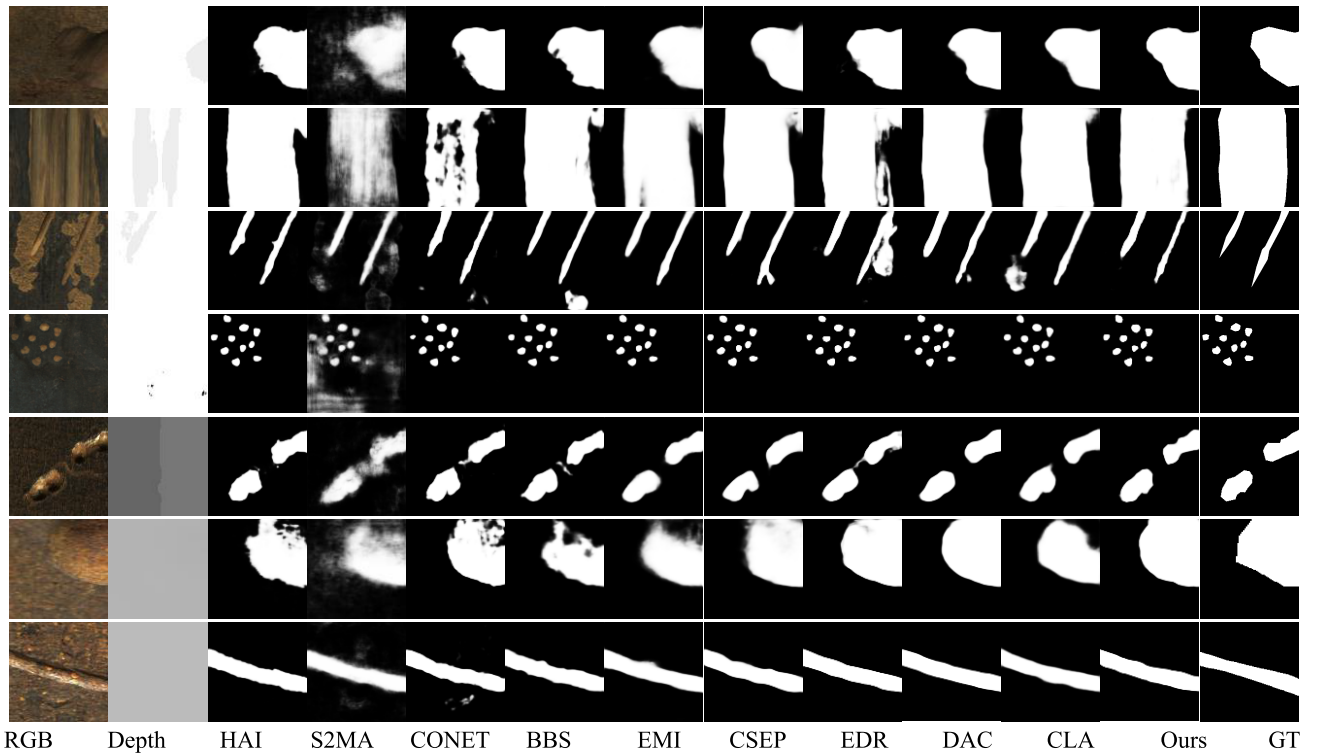


Fig. 6. Rail defect detection results of various evaluated methods.

TABLE IV
QUANTITATIVE PERFORMANCE OF ABLATION STUDIES

Models	NEU RSDDS-AU				
	$S_m \uparrow$	maxEm $\uparrow$	maxFm $\uparrow$	MAE $\downarrow$	fps(gpu) $\uparrow$
Backbone A	0.805	0.895	0.833	0.091	80.7
Backbone B	0.808	0.896	0.851	0.085	72.4
WO/BEM	0.821	0.911	0.867	0.076	62.5
WO/CEM	0.818	0.906	0.868	0.079	63.2
WO/MIM	0.823	0.912	0.869	0.074	63.1
Ours	0.836	0.926	0.881	0.064	60.3

is about 34 times smaller than the CLA model, the number of parameters is approximately 35 times less than the CLA model, and the FLOPs are approximately 30 times less than the CLA model. The proposed FHENet shows clear advantages in all the evaluation results.

Fig. 6 shows the rail defect detection results that verify the excellent performance of the proposed FHENet. Our proposal achieves defect detection in challenging scenes such as those with complex backgrounds and irregularities.

D. Ablation Studies

We conducted a series of ablation experiments to objectively demonstrate the importance and effectiveness of each core component of FHENet. The quantitative results are listed in Table IV.

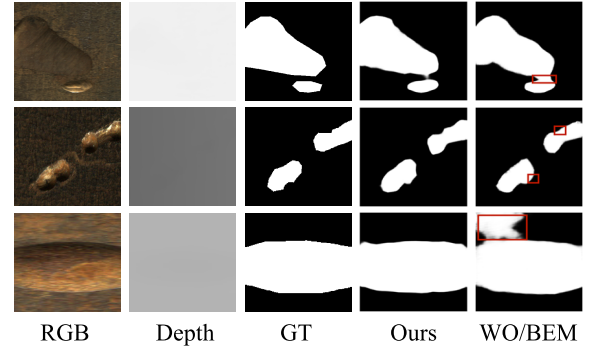


Fig. 7. Visual comparison results of WO/BEM and complete FHENet. The differences are marked by red boxes.

We adopted the two-stream encoder MobileNetV2 [7] to extract RGB and depth features. Then, the outputs of the last layer were concatenated as the final prediction result. This structure is denoted as backbone A. Moreover, we designed a hierarchical exploration network to realize cross-modal interaction.

This structure without any modules is denoted as backbone B.

Variant WO/BEM indicates FHENet discarding the BEM. The effectiveness of the BEM can be determined by examining Table IV. The BEM based on the maximum function and maximum pooling layer increases the clarity and accuracy of edges for defect detection, as shown in Fig. 7.

Variant WO/CEM indicates FHENet discarding the CEM. The usefulness of the CEM is indicated by the values listed in Table IV. The CEM comprehensively localizes defects,

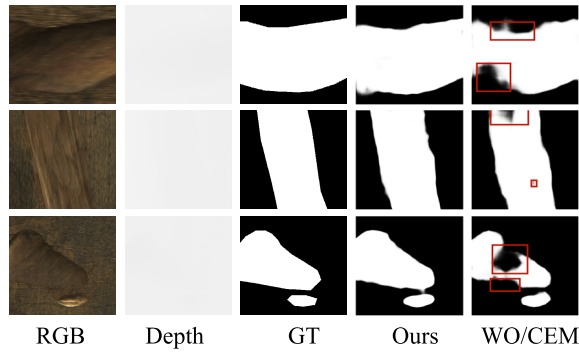


Fig. 8. Visual comparison results of WO/CEM and complete FHENet. The differences are marked by red boxes.

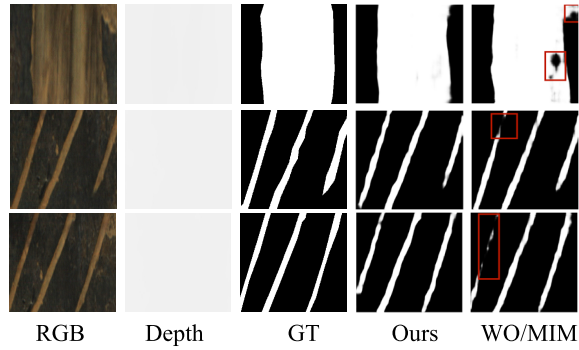


Fig. 9. Visual comparison results of WO/MIM and complete FHENet. The differences are marked by red boxes.

although it faces some complications in cases such as the similarity between background and foreground and irregularity of defects, as shown in Fig. 8.

Variant WO/MIM indicates FHENet without the MIM. Table IV shows that the MIM contributes to defect inspection. Similarly, the MIM achieves consistent semantic inspection when faced with multiple defects, as shown in Fig. 9.

According to Table IV, whenever we remove a module, the accuracy of the model decreases, but the inference speed of the model increases. This is because we replace the deleted module with a simple element-level addition operation, which reduces the complexity of the model computation process, reduces the number of parameters, and makes the computer itself more efficient at addition operations, which in turn increases the inference speed.

The aforementioned ablation experiments show the effect of every component of our network and validate the complete FHENet in comparison with incomplete variants.

E. Failure Cases

We proposed FHENet for identifying clear boundaries and complete objects while suppressing background noise interference. However, in some cases, like those shown in Fig. 10, FHENet does not perform well. In the first example, it is difficult to distinguish the defect from the background. In the second case, the quality of the RGB images is low, making FHENet unable to identify complete representative areas. In the last example of a cluttered scene, FHENet mistakenly identifies the background as a defect.

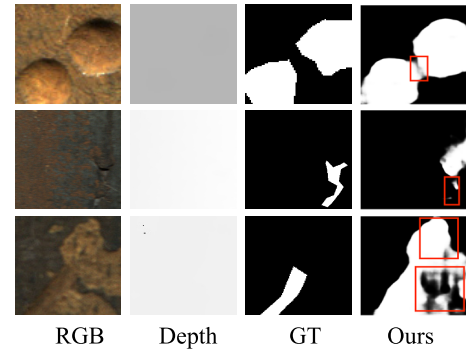


Fig. 10. Failure cases of proposed FHENet.

V. CONCLUSION

We proposed FHENet to obtain accurate boundaries of defect features and highlight information about salient defect regions using the novel BEM and hierarchical exploration. Multiple boundary supervision and deep supervision gradually refine the boundary and defect maps. Furthermore, FHENet establishes a lightweight architecture while ensuring high detection performance. Comprehensive experimental results demonstrated that the proposed FHENet outperformed ten state-of-the-art detection methods. As part of the future study, we will explore faster and more accurate methods to address rail defect detection using techniques such as knowledge distillation and graph convolution. In addition, we will use higher-resolution images, such as vector diagrams, to facilitate scene understanding and subsequent defect detection.

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